Stewart, Section 15.7, Exercise 23

Problem

(a) Express the volume of the wedge in the first octant that is cut from the cylinder $y^2 + z^2 = 1$ by the planes $y = x$ and $x = 1$ as a triple integral.

(b) Use either the Table of Integrals (on Reference Pages 6–10) or a computer algebra system to find the exact value of the triple integral in part (a).

Part (a)

Step 1. Name the surfaces that bound the solid

Four surfaces meet to make the boundary of the wedge.

- The surface $y^2 + z^2 = 1$ is a *circular cylinder*. Its axis is the x-axis, because x is the variable that is missing from the equation. Every cross-section of this cylinder that is perpendicular to the x-axis is a circle of radius 1 in a plane $x =$ constant.
- The plane $y = x$ is a vertical plane. It contains the z-axis and it cuts the cylinder along a curve.
- The plane $x = 1$ is a vertical plane. It closes the solid on one side.
- The three coordinate planes $x = 0$, $y = 0$, $z = 0$ bound the first octant.

The symbol E denotes the solid, that is, the set of points (x, y, z) inside the wedge. The symbol dV denotes the volume element. In the order of integration used below, $dV = dz\,dy\,dx$. The product $dz\,dy\,dx$ is the volume of a small box with sides dx, dy and dz.

Figure 1 shows the solid. The roof is the curved part of the cylinder, the floor is the triangle $z = 0$, and the two vertical planes are the faces $y = x$ and $x = 1$.

Step 2. Use the first octant and the two vertical planes

The first octant is the part of space where $x \geq 0$, $y \geq 0$ and $z \geq 0$. Inside this octant the cylinder keeps only the half where $z \geq 0$, so

$$z = \sqrt{1 - y^2},$$

and this expression is real only for $-1 \leq y \leq 1$. Together with $y \geq 0$ this gives $0 \leq y \leq 1$.

Next, the plane $y = x$ is a condition on x and y only. The wedge lies on the side of this plane where y is at most x, so $y \leq x$. The plane $x = 1$ gives $x \leq 1$. The two conditions together give

$$0 \leq y \leq x \leq 1.$$

This pair of inequalities describes a triangle in the xy-plane. Call this triangle T. It is the shadow, or the projection, of the solid on the plane $z = 0$.

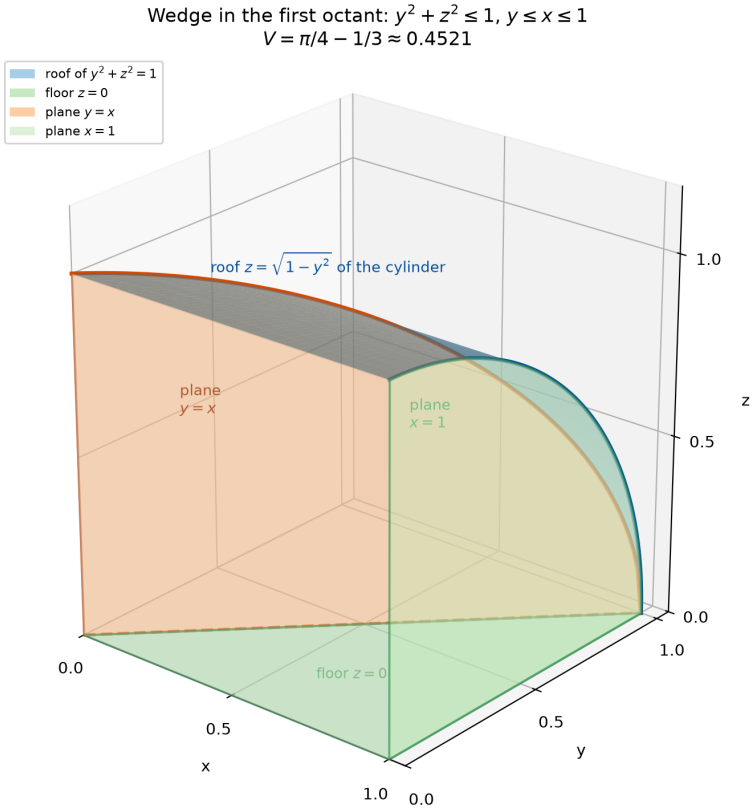


Figure 1: The wedge E of Exercise 23. The curved roof is the part of the cylinder $y^2 + z^2 = 1$. The floor is the triangle $0 \le y \le x \le 1$ in the plane $z = 0$. The plane $y = x$ is the orange face on the left, and the plane $x = 1$ is the quarter disk on the right. The brown curve is where the plane $y = x$ meets the roof, and the dark curve on the right is where the plane $x = 1$ meets the roof. The volume is $V = \pi/4 - 1/3 \approx 0.4521$.

Step 3. Write the solid as a set of inequalities

Put the three one-sided limits together. A point (x, y, z) is in E exactly when

$$E = \left\{ (x, y, z) \ : \ 0 \le x \le 1, \quad 0 \le y \le x, \quad 0 \le z \le \sqrt{1 - y^2} \right\}.$$

The last inequality is the cylinder condition $z^2 \le 1 - y^2$ with $z \ge 0$.

Step 4. Choose the order of integration

Fix a point (x, y) in the triangle T. The vertical line through this point is parallel to the z-axis. On this line the solid starts at the floor $z = 0$ and it stops at the curved roof $z = \sqrt{1 - y^2}$. The line meets E in one single interval, and neither limit depends on z. Therefore the order $dz\,dy\,dx$ is legal, and the inner integral runs from 0 to $\sqrt{1 - y^2}$.

The middle integral runs over y, and for fixed x the values of y go from 0 to x. The outer integral runs over x, and x goes from 0 to 1.

Step 5. Write the triple integral

With the volume element $dV = dz\,dy\,dx$ the volume is

$$V = \iiint_E 1\,dV = \int_0^1 \int_0^x \int_0^{\sqrt{1-y^2}} dz\,dy\,dx. \tag{1}$$

Equation (1) is the answer to part (a).

The same solid can be described with x and y exchanged. For $0 \le y \le 1$ the condition $y \le x \le 1$ gives x from y to 1, so

$$V = \int_0^1 \int_y^1 \sqrt{1-y^2}\,dx\,dy.$$

This second form is used in part (b), because the inner integral is then trivial.

Figure 2 shows the triangle T and one slice of the solid at a fixed value $y = y_0$. The slice is a rectangle with width $1 - y_0$ and height $\sqrt{1-y_0^2}$, and its area is $(1-y_0)\sqrt{1-y_0^2}$.

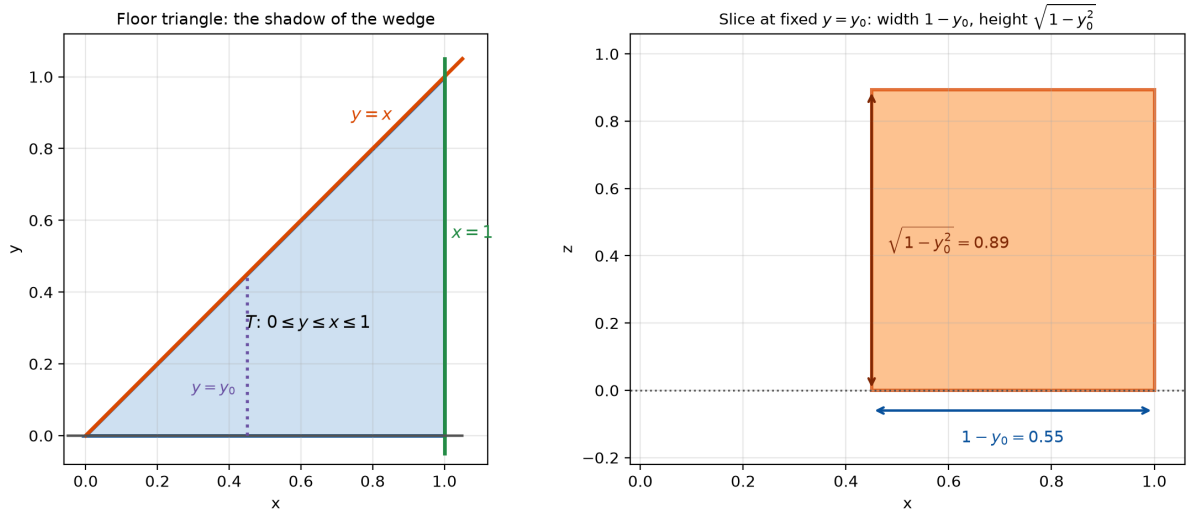


Figure 2: Left: the floor triangle T, where $0 \le y \le x \le 1$. Right: the slice of E at the fixed value $y = y_0 = 0.45$. The slice is a rectangle with width $1 - y_0 = 0.55$ and height $\sqrt{1-y_0^2} = 0.89$. The height is the same for every x in the slice, because the cylinder equation does not contain x.

Part (b)

Step 1. Integrate in z and in x

Start from equation (1). The inner integral is

$$\int_0^{\sqrt{1-y^2}} dz = \sqrt{1-y^2}.$$

The next integral is

$$\int_0^x \sqrt{1-y^2}\,dy,$$

and this one is not elementary in a short form. It is simpler to exchange x and y first, as in Step 5 of part (a). Then

$$V = \int_0^1 \left(\int_y^1 \sqrt{1-y^2}\, dx \right) dy = \int_0^1 (1-y)\sqrt{1-y^2}\, dy.$$

Step 2. Split the integral in two

The integrand is a difference, so the integral splits:

$$V = \int_0^1 \sqrt{1-y^2}\, dy - \int_0^1 y\sqrt{1-y^2}\, dy. \tag{2}$$

The first piece is a quarter of the area of the unit disk, so

$$\int_0^1 \sqrt{1-y^2}\, dy = \frac{\pi}{4}.$$

For the second piece put $u = 1-y^2$, so $du = -2y\, dy$ and $y\, dy = -du/2$. The limits change from $y = 0, 1$ to $u = 1, 0$, hence

$$\int_0^1 y\sqrt{1-y^2}\, dy = \int_1^0 \sqrt{u} \left(-\frac{du}{2} \right) = \frac{1}{2} \int_0^1 u^{1/2} du = \frac{1}{2} \cdot \frac{2}{3} = \frac{1}{3}.$$

Therefore

$$V = \frac{\pi}{4} - \frac{1}{3} = \frac{3\pi - 4}{12}.$$

Step 3. Check the same value with the table formula

The Table of Integrals gives

$$\int \sqrt{1-y^2}\, dy = \frac{y}{2}\sqrt{1-y^2} + \frac{1}{2} \arcsin y.$$

Its derivative is $\sqrt{1-y^2}$, so the formula is correct. With the upper limit $y = x$ and the lower limit $y = 0$ the inner integral is

$$\int_0^x \sqrt{1-y^2}\, dy = \frac{x}{2}\sqrt{1-x^2} + \frac{1}{2} \arcsin x.$$

Now integrate this expression in x on $[0, 1]$. The two terms give

$$\int_0^1 \frac{x}{2}\sqrt{1-x^2}\, dx = \frac{1}{2} \cdot \frac{1}{3} = \frac{1}{6}, \qquad \int_0^1 \frac{1}{2} \arcsin x\, dx = \frac{\pi}{4} - \frac{1}{2}.$$

The second value follows from $\int \arcsin x\, dx = x \arcsin x + \sqrt{1-x^2}$, which gives $\pi/2 - 1$ at the upper limit and 1 at the lower limit. Adding the two terms,

$$V = \frac{1}{6} + \left(\frac{\pi}{4} - \frac{1}{2} \right) = \frac{\pi}{4} - \frac{1}{3}.$$

Both routes give the same answer. Figure 3 shows the two pieces of the answer as areas under the two curves.

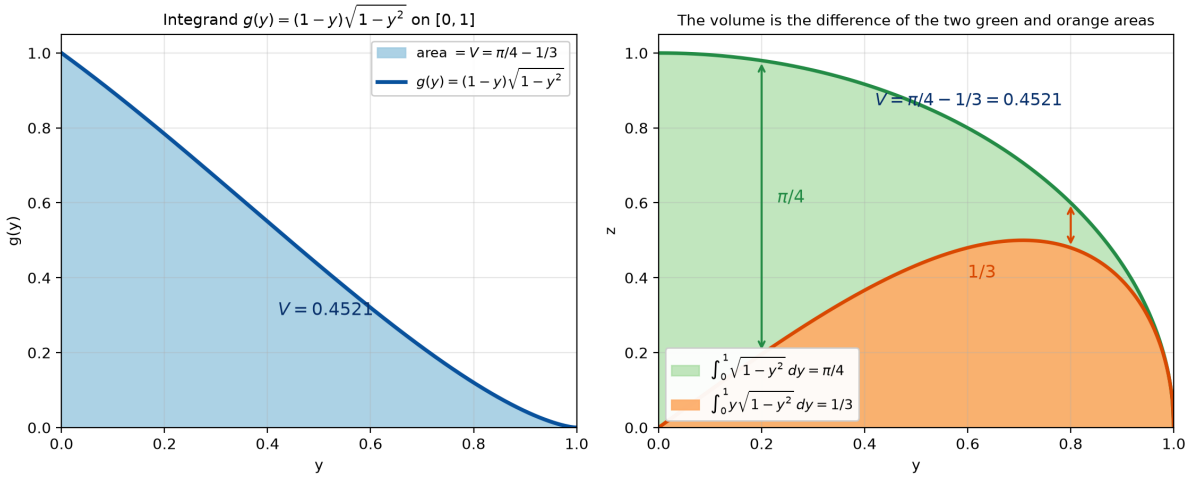


Figure 3: Left: the integrand $g(y) = (1-y)\sqrt{1-y^2}$ on $[0, 1]$. The area under the curve is the volume. Right: g is the difference of $\sqrt{1-y^2}$ and $y\sqrt{1-y^2}$. The green area is $\pi/4 = 0.7854$ and the orange area is $1/3 = 0.3333$, so the volume is 0.4521.

Step 4. Check the value with numbers and with bounds

The exact value is

$$V = \frac{3\pi - 4}{12} = 0.452064830064\ldots$$

Three numerical methods in `part1_step3.py` and `part2_step2.py` confirm this value. Table 1 lists the results.

Method	Value	Error
Midpoint rule on T, 4000^2 cells	0.4520648344	4.3×10^{-9}
Simpson rule in y, 4000 panels	0.4520648301	2.0×10^{-11}
Monte Carlo, 4×10^6 points	0.4521387500	7.4×10^{-5}

Table 1: Numerical values of the volume. The last error is inside three standard deviations of the Monte Carlo estimate, and that standard deviation is 2.5×10^{-4}. The two deterministic rows agree with the exact value to better than 10^{-8}.

Two more checks fix the size of the answer. The solid fits inside the box $0 \le x \le 1$, $0 \le y \le 1$, $0 \le z \le 1$, which has volume 1, so $V < 1$. The solid is also a part of the quarter cylinder $0 \le y \le 1$, $0 \le z \le \sqrt{1-y^2}$, $0 \le x \le 1$, whose volume is $\pi/4 = 0.7854$. The plane $y = x$ removes part of that quarter cylinder, so $0 < V < \pi/4$. The value 0.4521 is 0.58 of $\pi/4$, and it has the right size for a wedge that keeps a little more than half of the quarter cylinder.

Step 5. State the answer

$$\boxed{V = \frac{\pi}{4} - \frac{1}{3} = \frac{3\pi - 4}{12} \approx 0.4520648301}$$

How to solve problems of this kind

1. Find the axis of the cylinder. The variable that is missing from the equation is the direction of the generator lines. Here $y^2 + z^2 = 1$ has no x, so the axis is the x-axis and every cross-section perpendicular to that axis is a circle.

2. Use the first octant first. Set $x \geq 0$, $y \geq 0$, $z \geq 0$. This removes half of the cylinder and it makes $z = \sqrt{1 - y^2}$ a single-valued function, so no absolute value is needed.

3. Read the two vertical planes as limits on the shadow. The plane $y = x$ gives $y \leq x$ and the plane $x = 1$ gives $x \leq 1$, so the shadow is the triangle $0 \leq y \leq x \leq 1$ in the plane $z = 0$.

4. Read the inner limits from the curved surface. Put the cylinder equation in the form $z = \sqrt{1 - y^2}$, and put the variable of the axis outermost. Here the inner integral in z runs from the floor $z = 0$ to the roof $z = \sqrt{1 - y^2}$.

5. Choose the order that makes the inner integral easy. The order $dz\,dy\,dx$ gives a hard y-integral, but exchanging x and y first gives the factor $1 - y$ and the table formula applies. A legal exchange never changes the value.

6. Split the integrand when it is a difference. Here $(1 - y)\sqrt{1 - y^2}$ splits into $\sqrt{1 - y^2}$ and $y\sqrt{1 - y^2}$. The first is a quarter disk and the second is a power of $u = 1 - y^2$.

7. Check the answer three ways. Use a second order of integration, a numerical quadrature, and the size of a box that contains the solid.